Appendix of COPunD

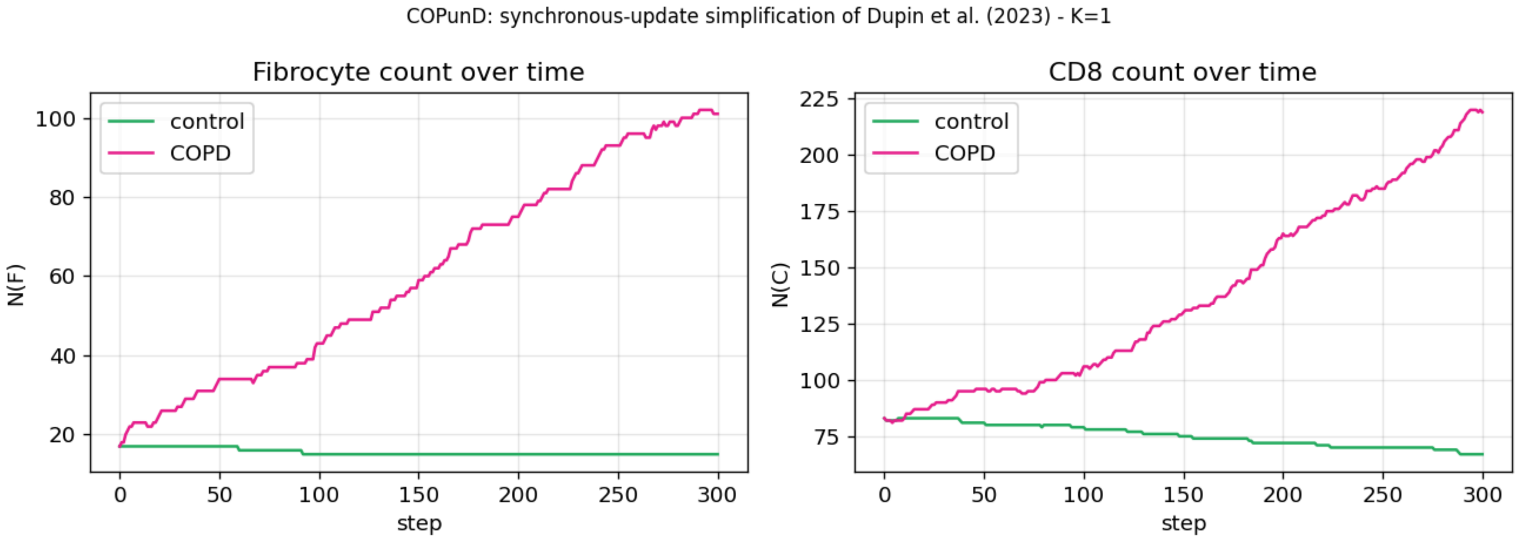

Figure 1. Fibrocytes and CD8 counts over time during 12 days.

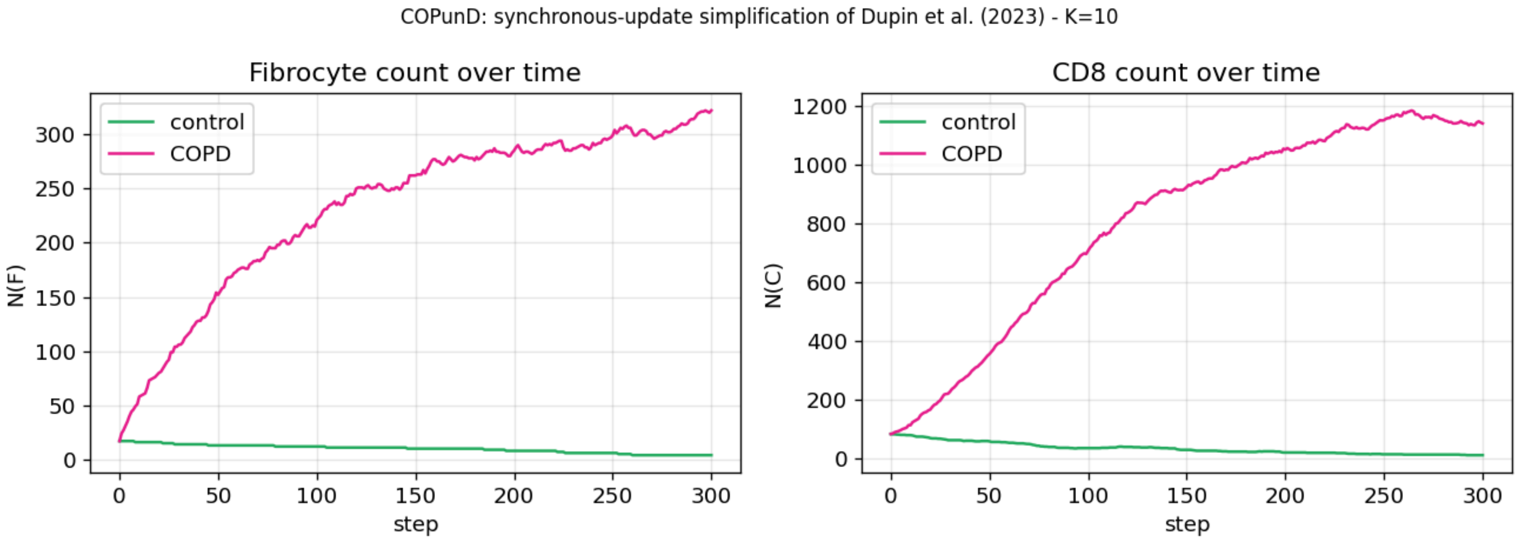

Figure 2. Fibrocytes and CD8 counts over time during 125 days.

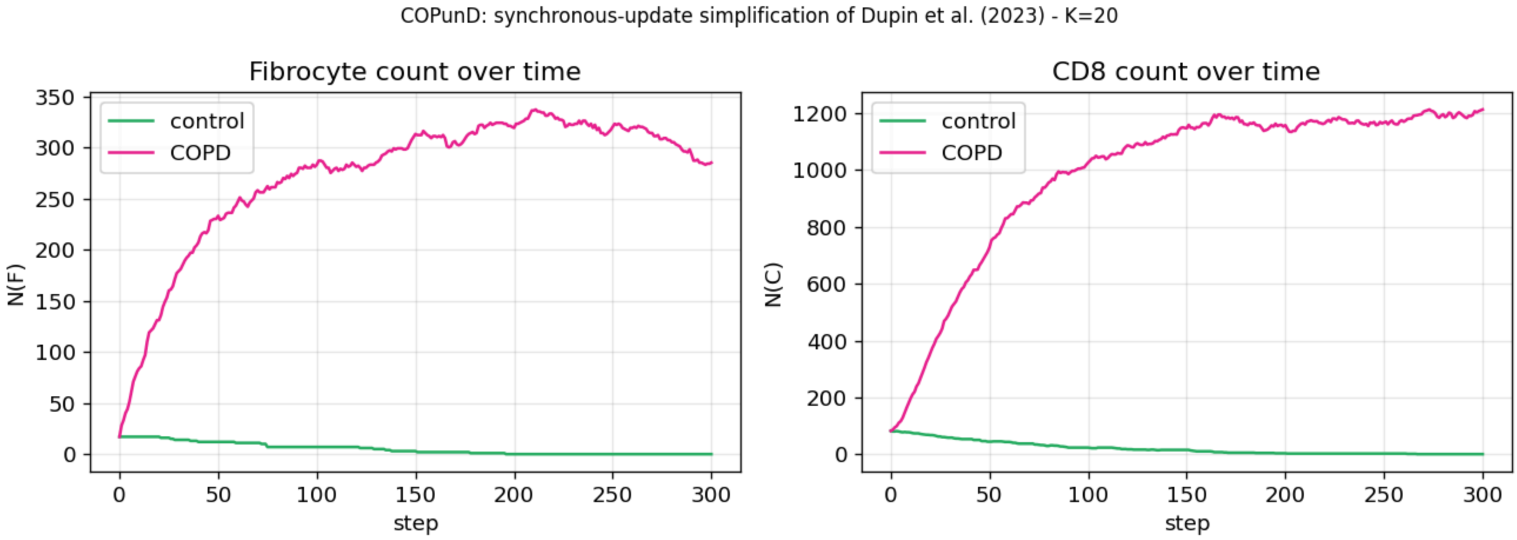

Figure 3. Fibrocytes and CD8 counts over time during 250 days.

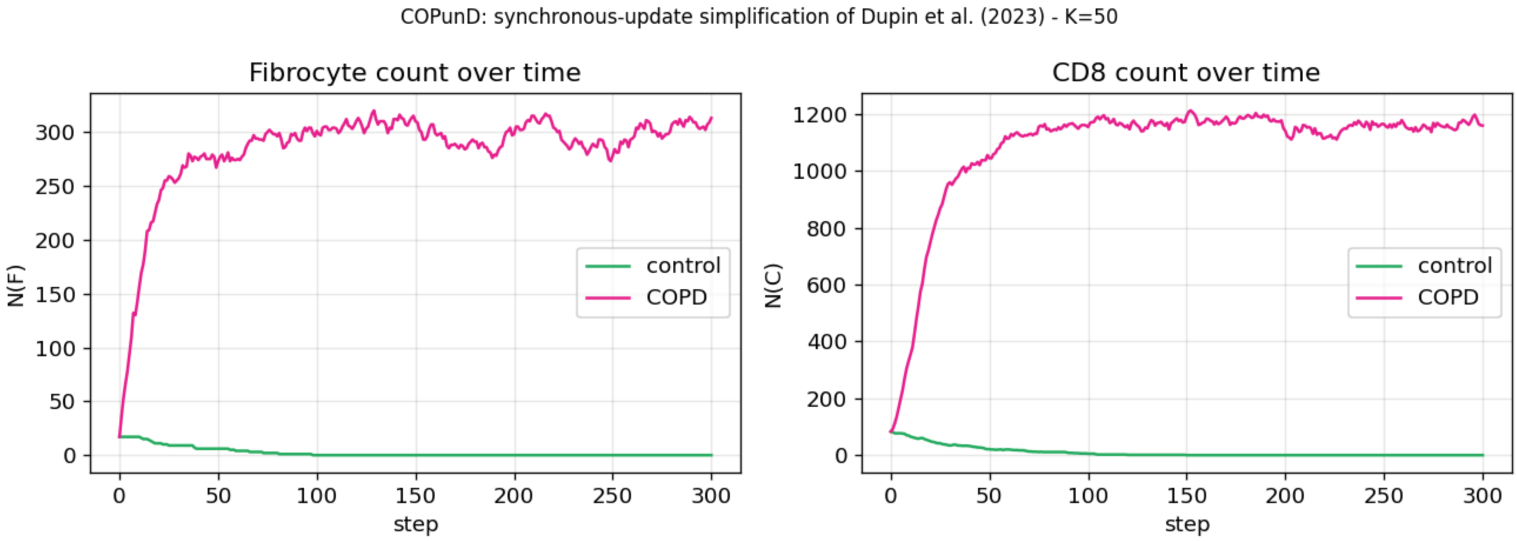

Figure 4. Fibrocytes and CD8 counts over time during 625 days.

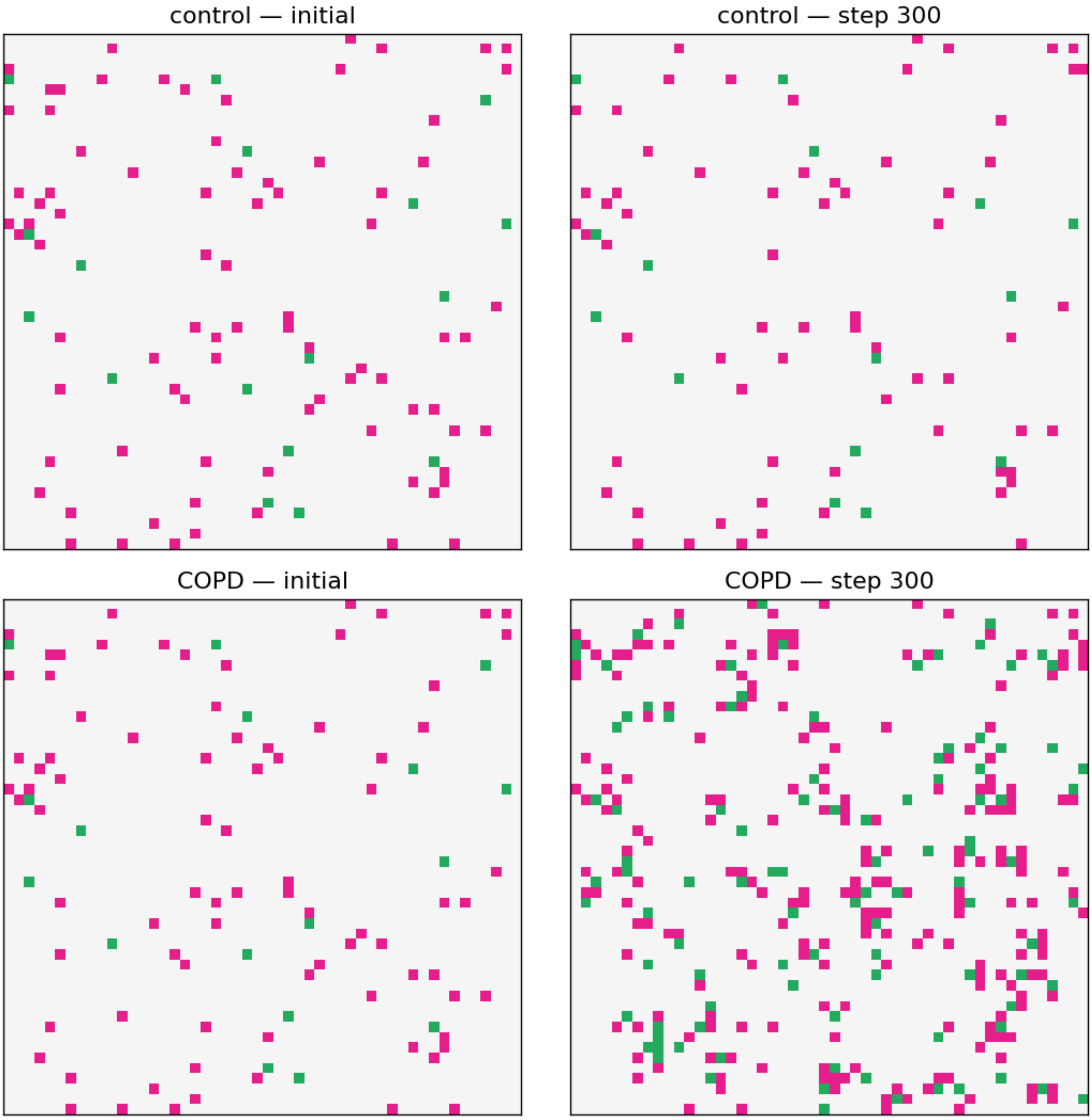

Figure 5. Snapshots initial & final simulation states (Control VS COPD), 12 days.

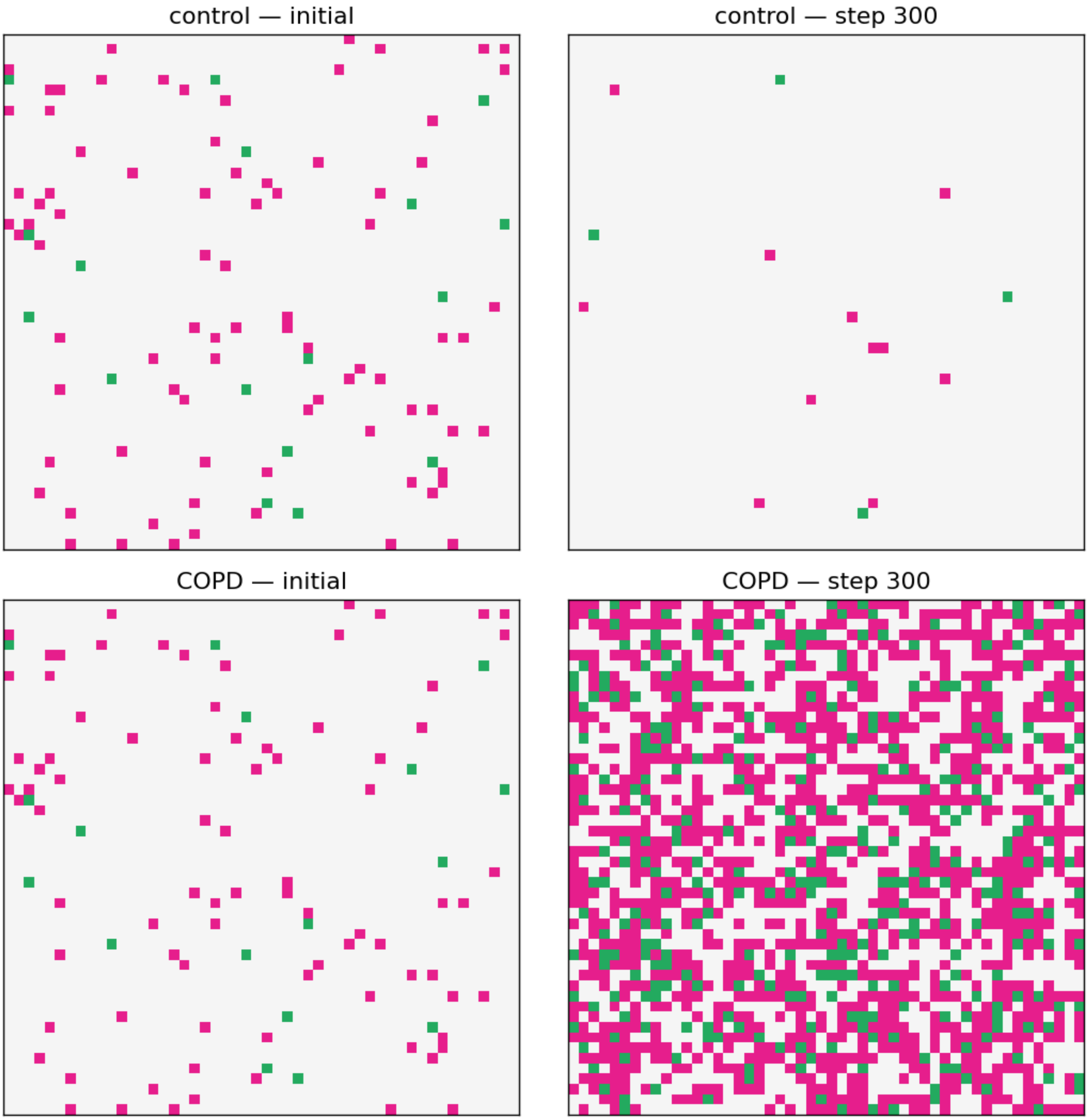

Figure 6. Snapshots initial & final simulation states (Control VS COPD), 125 days.

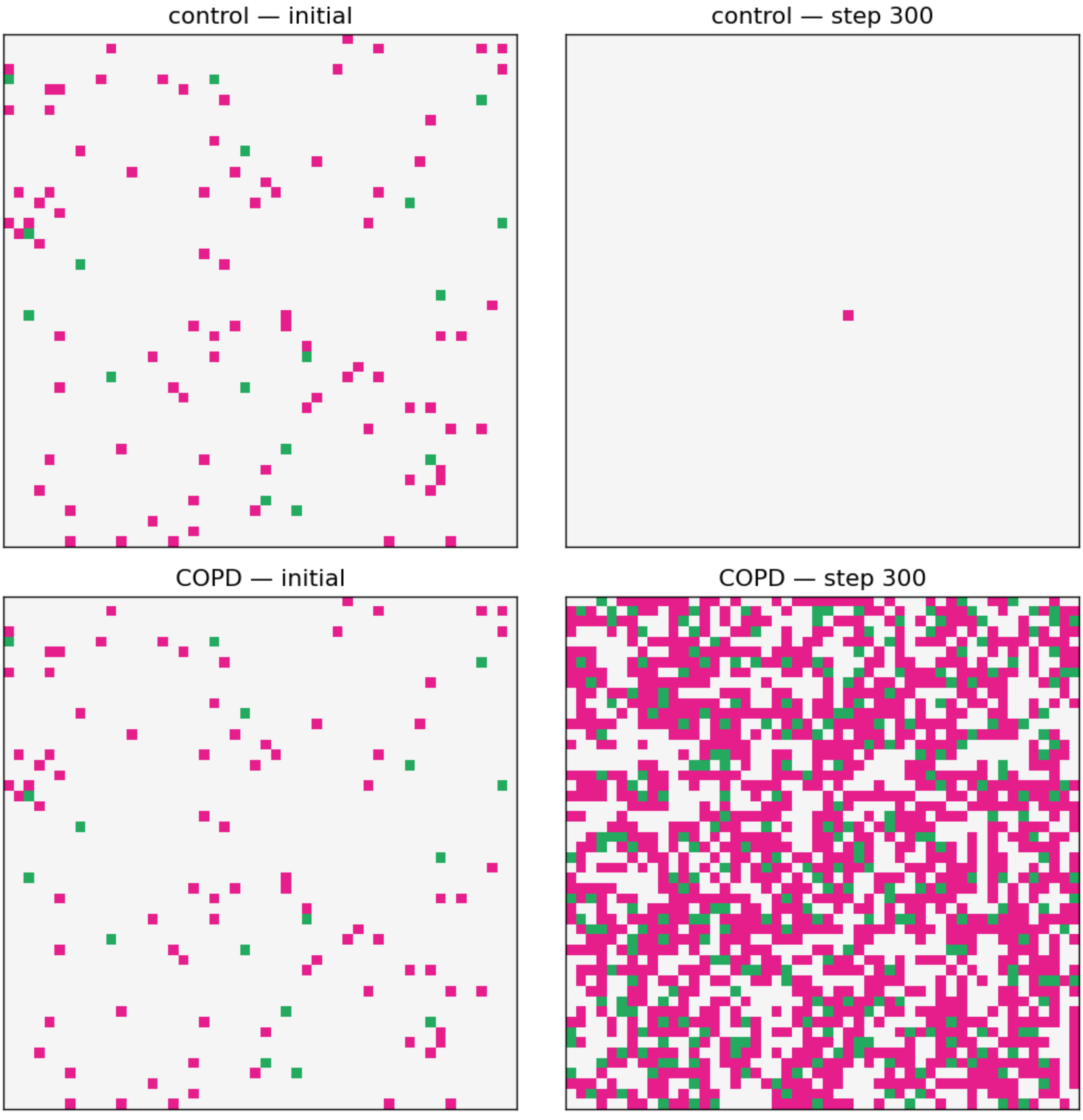

Figure 7. Snapshots initial & final simulation states (Control VS COPD), 250 days.

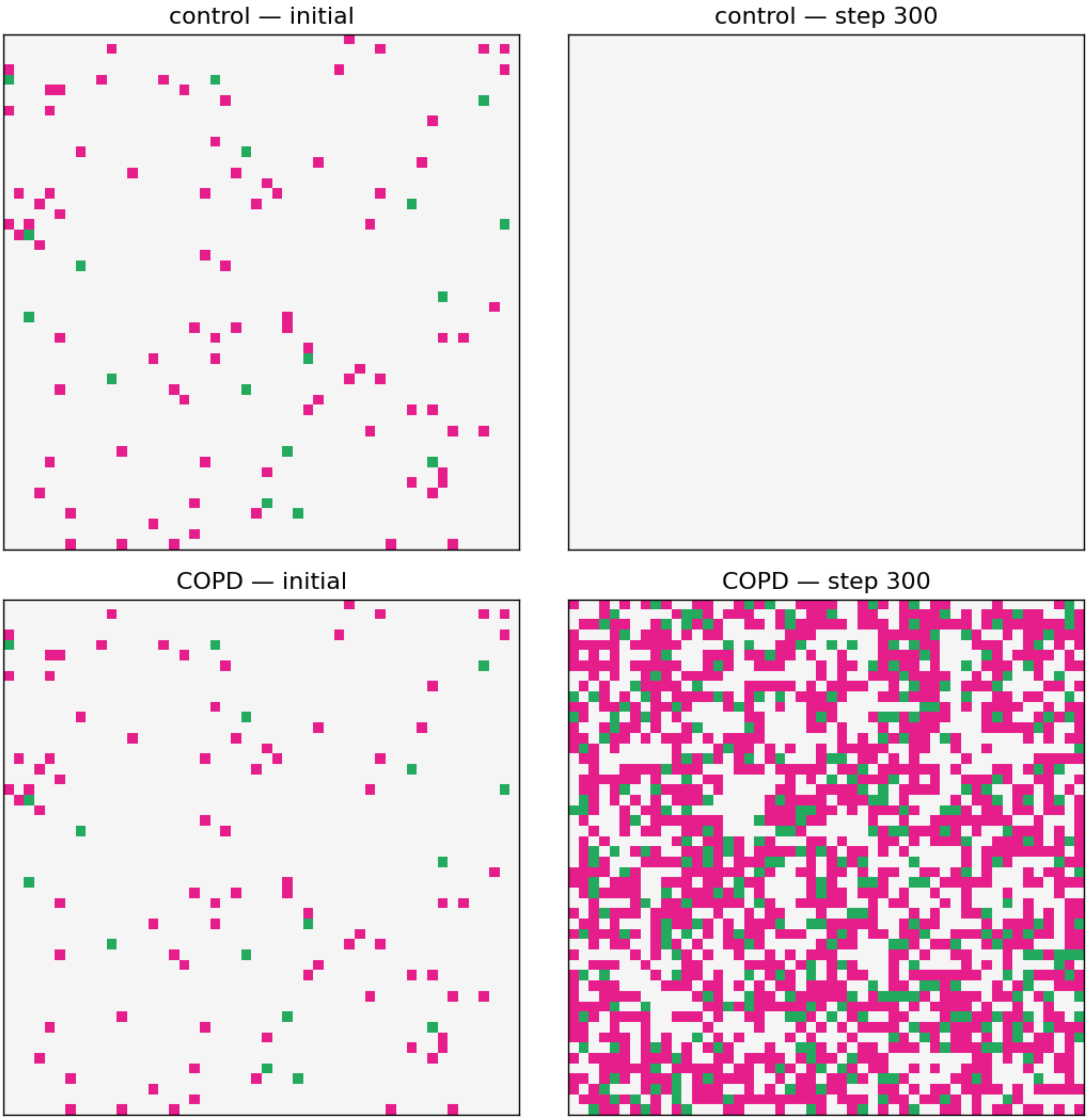

Figure 8. Snapshots initial & final simulation states (Control VS COPD), 625 days.

Aspect	*Dupin et al. (2023) model*	*COPunD model*	*Limitation introduced*
Biological scope	Interactions between fibrocytes and CD8+ T lymphocytes in the peribronchial area, while acknowledging that real lung tissue contains many other structural and immune cells.	Only three states: Empty, Fibrocyte, and CD8. Other biological components are grouped into the empty state.	The model does not represent epithelial cells, smooth muscle cells, extracellular matrix effects, or other immune cells. This reduces biological realism.
Tissue geometry	Lamina propria as a crown-shaped region surrounding the bronchial lumen, using a biologically realistic geometry.	A simple square two-dimensional grid.	The simplified grid does not reproduce the real bronchial geometry or boundary effects.
Probabilities	Many biologically estimated probabilities for death, proliferation, migration, stable infiltration, and exacerbation-related infiltration.	Our model uses fewer probabilities: fibrocyte death, CD8 death, fibrocyte infiltration, and CD8 birth near fibrocytes.	The probability structure is less detailed, so the model is qualitative rather than quantitatively calibrated.
CD8 proliferation	CD8 proliferation depends on biological parameters and local conditions, especially the presence of fibrocytes and crowding effects.	In our model, CD8 cells can only appear if at least one fibrocyte is present nearby.	This captures the main inflammatory interaction but removes finer mechanisms such as thresholds, contact inhibition, and detailed proliferation rules..
Fibrocyte infiltration	Distinction between stable-state and exacerbation infiltration, particularly in COPD conditions.	Our model activates fibrocyte infiltration only in COPD mode.	The model does not separately represent stable infiltration and exacerbation events.
Migration	Migration depends on transition probabilities between neighbours and local attraction functions, with fibrocytes attracted by CD8 signals and movement influenced by local density.	Our model uses a simplified mobility rule. Fibrocytes move more than CD8 cells, and movement decreases when same-type neighbors are present.	Migration is not a full biological chemotaxis model. It mainly produces clustering behavior rather than accurately reproducing cell displacement.
Update process	Update cells through sub-time steps, where cells are selected and processed individually.	Our model uses the synchronous update mechanism of the course cellular automaton framework.	Synchronous updates introduce artifacts and deviate from the original cell-by-cell dynamics.
Number of parameters	Many parameters, including death, proliferation, infiltration probabilities, attraction functions, thresholds, and time-scale parameters.	Uses a reduced parameter set: *p_F_die, p_C_die, p_F_infil, p_C_birth, P_MOVE_F, P_MOVE_C, and K_TIME.*	Fewer parameters make the model easier to implement and interpret, but less biologically accurate.
Purpose	The paper aims to build a biologically informed and mathematically analyzed model of COPD-related cell interactions.	Our model aims to reproduce the main qualitative behavior of COPD-like inflammation in a simplified computational setting.	The model should be interpreted as an educational and qualitative simplification, not as a predictive biomedical model.

Table 1. Limitations & differences between Dupin et al. model and COPunD model.